

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 5 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCVC502 - GEOTECHNICAL ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Terzaghi's effective stress principle states that:
 - A) Total stress controls soil strength
 - B) Neutral pressure increases soil shear strength
 - C) Stress carried by soil skeleton governs strength and deformation
 - D) Water pressure increases bearing capacity
 - 2 Terzaghi's effective stress equation is:
 - A) $\sigma = \sigma' + u$
 - B) $\sigma' = \sigma + u$
 - C) $\sigma' = \sigma - u$
 - D) $\sigma = \sigma' - u$
 - 3 Soil moisture refers to:
 - A) Water around soil grains only
 - B) Water filling soil voids
 - C) Water chemically absorbed by minerals
 - D) Surface water only
 - 4 Sand replacement method is best suited for:
 - A) Very hard soil
 - B) Very soft saturated soil
 - C) Irregular and cohesionless soil
 - D) Smooth cohesive soil
 - 5 Atterberg limits apply mainly to:
 - A) Coarse-grained soil
 - B) Gravel only
 - C) Fine-grained soils (clays & silts)
 - D) Sand and gravel
 - 6 High plasticity index indicates:
 - A) High sand content
 - B) High compressibility & swelling
 - C) Low clay content
 - D) High density index

- 7 Quick sand condition occurs when:
 - A) Seepage pressure = effective stress
 - B) Total stress becomes zero
 - C) Permeability becomes infinite
 - D) Seepage is horizontal
- 8 Equipotential lines in flow net represent:
 - A) Direction of flow
 - B) Equal seepage velocity
 - C) Equal hydraulic potential drop
 - D) Total stress variation
- 9 Shear strength of saturated clays under UU conditions depends mainly on:
 - A) Effective stress
 - B) Total stress
 - C) Cohesion only
 - D) Angle of friction only
- 10 Mohr's circle helps determine:
 - A) Pore pressure
 - B) Principal stresses & failure plane
 - C) Permeability
 - D) Consolidation properties
- 11 Consolidation occurs due to:
 - A) Change in air content
 - B) Expulsion of pore water
 - C) Rearrangement of particles without drainage
 - D) Elastic rebound
- 12 Active earth pressure decreases when:
 - A) Wall moves away
 - B) Wall is rigid
 - C) Soil is partly saturated
 - D) Cohesion increases

Q2. Solve the following.

- A) Explain the three-phase system of soil with a neat diagram. 6
- B) The water table in a deposit of sand 8 m thick, is at a depth of 3 m below the surface. Above the water table, the sand is saturated with capillary water. The bulk density of sand is 19.62 kN/m^3 . Calculate the effective pressure at 1 m, 3 m and 8 m below the surface. Hence plot the variation of total pressure, neutral pressure and effective pressure over the depth of 8 m. 6

Q3. Solve the following.

- A) Explain the Following terms: 6
 - 1) Voids Ratio 2) Porosity 3) Degree of saturation
 - 4) Water Content 5) Specific Gravity 6) Air Content
- B) The Atterberg limits of a clay soil are: liquid limit 52%, plastic limit 30% and shrinkage limit 18%. If the specimen of this soil shrinks from a volume of 39.5 cm^3 at the liquid limit to a volume of 24.2 cm^3 at the shrinkage limit, calculate the true specific gravity.

Q4. Solve Any Two of the following.

- A) State Darcy's Law and discuss the factors affecting permeability. 6

- B) Describe Falling Head Permeability Test with proper diagram. 6
- C) Calculate the coefficient of permeability of a soil sample, 6 cm in height and 50.6 cm² in cross-sectional area, if a quantity of water equals to 430 ml passed down in 10 minutes, under an effective constant head of 40 cm. On oven-drying, the test specimen has mass of 498 g. Taking the specific gravity of soil solids as 2.65, calculate the seepage velocity of water during the test.

Q5. Solve Any Two of the following.

- A) Enlist & explain in detail the types of shear test based on drainage conditions. 6
- B) A vane, 10 cm long and 8 cm in diameter, was pressed into soft clay at the bottom of a bore hole. Torque was applied and gradually increased to 45 N-m when failure took place. Subsequently, the vane rotated rapidly so as to completely remould the soil. The remoulded soil was sheared at a torque of 18 N-m. Calculate the cohesion of the clay in the natural and remoulded states and also the value of the sensitivity.
- C) Explain Coulomb's shear strength theory and the concept of the failure envelope. 6
How do stress conditions and soil type influence the failure envelope?

Q6. Solve Any Two of the following.

- A) State and explain the assumptions in 'Terzaghi's theory of one-dimensional consolidation. 6
- B) Explain active and passive earth pressure concepts? 6
- C) Describe the comparison between Rankine and Coulomb's earth pressure theories. 6

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